

Cartesian product:

$$A = \{1,2,3\}, B=\{a,b\}$$

$$A \times B = \{(1,a), (2,a), (3,a), (1,b), (2,b), (3,b)\}$$

$$B \times A = \{(a,1),(b,1),(a,2),(b,2),(a,3),(b,3)\}$$

Number of elements / cardinality:

$$|A| = 3$$

$$|B| = 2$$

$$|A \times B| = |A| * |B| = 3 * 2 = 6$$

Relations:

$$a = \{1,4,5\}$$

$$b = \{1,2,6\}$$

$$R = \{(a,b) \mid a > b\}$$

We want to get the Cartesian product where the $a > b$ relation applies

$$a = 1$$

$$1 = 1 \rightarrow \emptyset$$

$$1 < 2 \rightarrow \emptyset$$

$$1 < 6 \rightarrow \emptyset$$

$$a = 4$$

$$4 > 1 \rightarrow \{(4,1)\}$$

$$4 > 2 \rightarrow \{(4,2)\}$$

$$4 < 6 \rightarrow \emptyset$$

$$a = 5$$

$$5 > 1 \rightarrow \{(5,1)\}$$

$$5 > 2 \rightarrow \{(5,2)\}$$

$$5 < 6 \rightarrow \emptyset$$

$$R = \{(4,1),(4,2),(5,1),(5,2)\}$$

$$R = \{(a,b) \mid a = 4\}$$

$$1 \neq 4 \rightarrow \emptyset$$

$$4 = 4$$

$$5 \neq 4 \rightarrow \emptyset$$

$$\{(4,1), (4,2), (4,6)\}$$

$$R = \{(4,1),(4,2),(4,6)\}$$

$$R = \{(2,4),(2,6),(4,7),(6,9),(7,7)\}$$

Domain: what numbers are present on the first coordinate of a point (x,y)

$$D_R = \{2,4,6,7\}$$

Range: what numbers are present on the second coordinate

$$R_R = \{4,6,7,9\}$$

Inverse of R:

$$R^{-1} = \{(4,2),(6,2),(7,4),(9,6),(7,7)\}$$